Introduction to Psycholinguistics

Unit 3

Spoken Word Recognition

(Contd.)

Contact and Activation

- Once pre-lexical analysis has broken the speech signal down into usable units, the next stage is contact, establishing links between the output of that pre-lexical analysis and the word-forms that are actually stored in the mental lexicon.
- This kind of mapping, from the raw input toward the lexicon, is **bottom-up processing**; information flows from the signal upward, rather than being driven by expectations from above.
- Many models argue that this initial contact stage is based purely on automatic bottom-up processing.
- Other models additionally claim a role here for top-down processing, for instance, using sentence context to pre-select words from a particular area of meaning before the bottom-up signal even arrives

Parallel Contact vs. Serial Search: Two Competing Architectures

Parallel models of lexical processing

- These are characterized by the idea that the input can make **simultaneous contact with more than one stored word at once.**
- Multiple candidate words all get contacted together, based on shared input.

Serial search models

- Here, candidate words are considered **one at a time**, in sequence, rather than all together.

The Key Evidence: Zwitserlood's (1989) Cross-Modal Priming Study

- What is cross-modal priming?



Papaya



Response (either during or immediately)

- The logic is: if hearing the prime has activated a related word in the listener's mind, then a visually presented target related to that activated word should be recognized/responded to faster.
- This speeding-up effect is called facilitation.

- Design of the experiment (originally run in Dutch)

Test prime: **captain**

Control prime: justice (unrelated to captain, used as a baseline comparison)

- The critical trick: the test prime captain has what's called a "partner word": a word that starts with exactly the same sequence of phonemes but is a different word entirely: **capital**.
- Crucially, this partner word (capital) was never actually spoken in the experiment at all; its relevance is purely that it is a phonological "competitor" that could still be forming in the listener's mind while they're hearing the early portion of captain, since /kæp.../ is genuinely ambiguous between the two.

- Three types of visual target were used, each combined with both the test prime and the control prime:
- **Target 1**: ship: semantically related to the test prime, captain.
- **Target 2**: money: semantically related to the unspoken partner word, capital (a captain doesn't relate to money, but "capital" clearly does).
- **Target 3**: lamp: the baseline/control target, related to neither captain nor capital.
- The critical manipulation: timing of the target relative to the prime:
  - Early condition: the visual target was presented part-way through the spoken prime, specifically, at a point that includes exactly the material shared by captain and capital, which (for this pair) is right at the end of the shared /kæp/ portion, i.e., before the whole prime word has actually been heard.
  - Late condition: the visual target was presented only after the entire prime word had been heard in full.

The results:

- In the Early condition: **facilitation was found** for targets related to both the test prime and its unspoken partner word. In our example, *that means both ship (related to captain) and money (related to capital) were responded to more rapidly and more accurately after hearing the early fragment of captain, compared to after hearing the equivalent early fragment of the unrelated control prime justice*. The control target, lamp, showed no such facilitation in either condition- confirming the effect is specific to genuine semantic relatedness, not some general speeding-up artifact.
- In the Late condition- once the entire prime word had been heard **- only ship** (related to captain, the word that was actually said) **showed facilitation**. *The earlier facilitation effect for money (related to the never-spoken capital) had completely disappeared by this point*.

What this tells us, stated as the core theoretical conclusion:

- During the early stages of processing the spoken word captain, both captain itself and the phonetically similar but ultimately incorrect word capital have been contacted simultaneously- and, importantly, words related in meaning to both of these contacted candidates get primed at that early stage.
- In other words, **the initial contact stage is parallel**: multiple words matching the bottom-up input all get contacted and activated together, not just the one word that will eventually turn out to be correct. As more of the input arrives and the signal diverges from capital, that competitor drops out of contention, and only the correct word's associates continue to show priming.

The Word-Initial Cohort: Only Beginnings Matter

- A further, important refinement comes from experiments using rhyming word pairs, which help isolate **exactly which part of a word drives this contact/priming effect**.
- Worked contrast: the word **bee** is primed by **honey**, but it is **NOT** primed by **money**, even though money actually shares most of the same sounds as honey (they rhyme - both end in "-oney").
- The generalization: it is specifically *the overlap of the initial portion of a word that drives priming/contact*, *not just any portion of the word*, however much phonetic material is shared overall.

Why this makes good design sense

- Functionally: it’s logical for the beginning sounds of a word to carry the primary responsibility for making contact with entries in the mental lexicon, simply because, in real-time speech processing, we hear the beginning of a word first, the system has to start working with whatever information arrives earliest.
- Terminology, the ‘**word-initial cohort**’: the set of words that get contacted this way, purely on the basis of sharing the same initial sequence of sounds, is referred to as a word's word-initial cohort.

The Cohort Model (Marslen-Wilson & Welsh, 1978; Marslen-Wilson, 1987)

The word-initial cohort concept is the defining feature of an influential model of spoken word recognition called the **Cohort Model**, a parallel, primarily bottom-up approach.

- Once the initial sounds of a word have been heard, all words in the listener's mental lexicon that share that same initial sound sequence become contacted simultaneously.
- Once this initial contact has happened, the lexical entries for all of these cohort members become activated, which in turn makes available further stored information about each of these words (their meanings, grammatical properties, etc.)

How the cohort narrows over time:

- Zwitserlood's experiment (mentioned previously) also demonstrates that as the speech input continues to unfold, further bottom-up information about the upcoming sounds becomes available, and at that point, some of the originally contacted cohort members stop matching the input and consequently '**fall out of contention**.'
- A useful directional metaphor, borrowed from reading: this process is sometimes described (at least for languages like English that are read left-to-right) as involving left-to-right processing of spoken language, the cohort narrows down progressively as the word unfolds from beginning to end.
- By the very end of the spoken word captain, it has become completely clear that the word is captain and not capital, and so, at that final point, only ship remains primed; money is no longer primed at all.

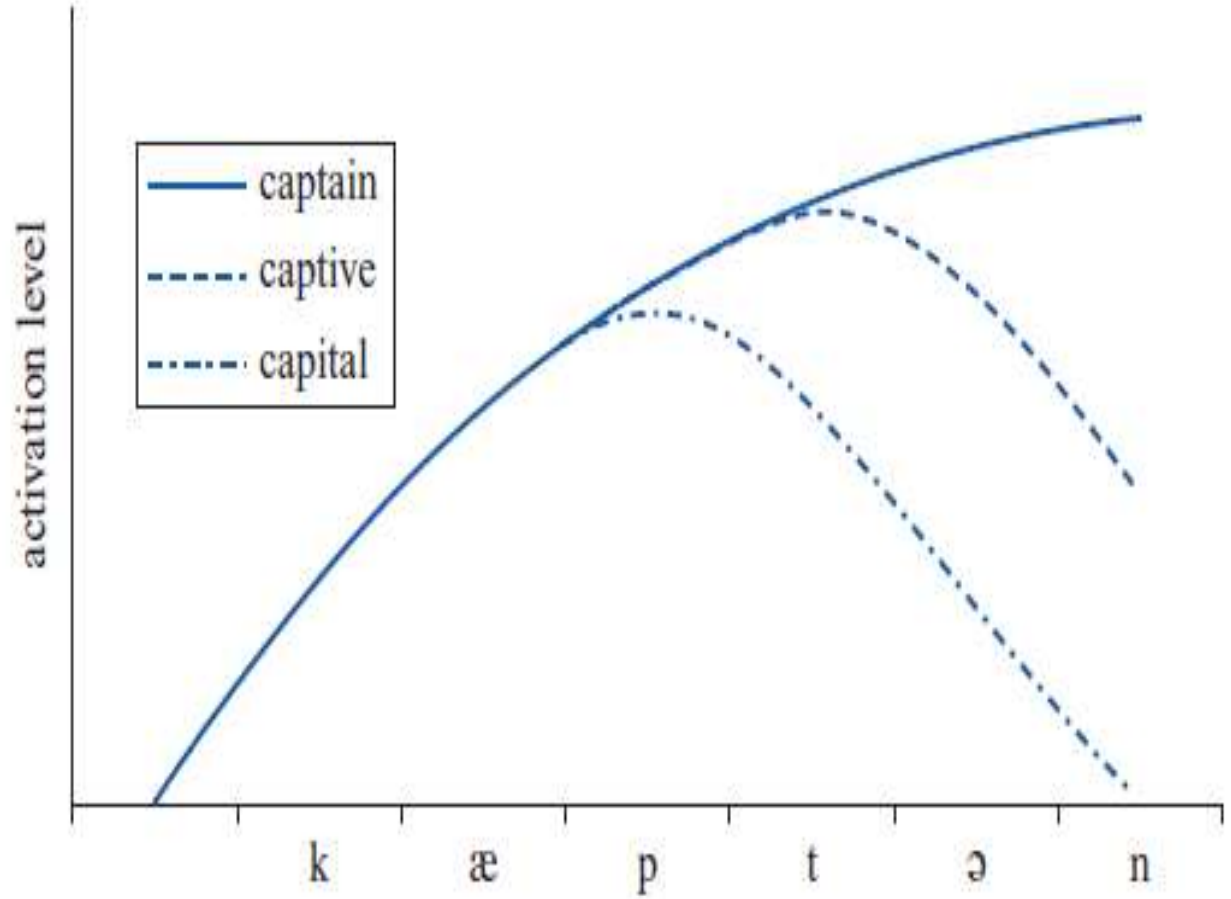


Figure 8.1
Illustration of changing activation levels of word entries as more of a word (*captain*) is heard.

From ‘On/Off’ to Graded Activation Levels

- Early versions of the Cohort Model made a fairly simple, binary assumption: a candidate word was treated as either **fully available** or **fully unavailable**, depending simply on whether its sounds currently matched the incoming input.
- This binary, all-or-nothing view was later abandoned. It became widely recognized that lexical entries in the mental lexicon are not simply switched ‘**on**’ or ‘**off**’, instead, they take on **graded levels of activation** (Marslen-Wilson, 1987).

How graded activation works:

- The more input there is that supports/matches a particular lexical candidate, the higher that entry's level of activation becomes.
- Worked example, laid out over time: consider the activation levels of captain, captive, and capital as the word captain is heard, sound by sound
- As the very first few sounds (/kæp/) arrive, the activation levels of both captain and capital rise together, since both are still fully consistent with the input so far.

- Once the sound pattern of the actual input starts to diverge from what would be expected for capital, that word's activation begins to fall away, while the activation of captain (the word actually being said) continues to rise.
- This graded rise-and-fall pattern is illustrated in the text's Figure 8.1, tracking the changing activation levels of captain, captive, and capital as more and more of the word captain (sound sequence /k æ p t ə n/) is heard over time.

Why Graded Activation Is a Better Model: The Mispronunciation-Recovery Advantage

The key theoretical advantage of an activation-based (graded) model over a strict all-or-nothing model: it much more naturally allows for partial matches, and crucially, it allows for recovery from mispronunciation or from unexpected/non-standard pronunciations- something a rigid binary on/off system would struggle to handle gracefully.

- example: consider a speaker who pronounces "student" as "shtudent“- an increasingly common pronunciation noted in New Zealand English.
- Under a graded-activation account: the initial "sh" sound partially activates the similar phoneme /s/ (rather than failing to match at all), and then the subsequent incoming sounds (/tju.../) continue to progressively increase the activation of the word "student" (which we can suppose is stored in the mental lexicon in its standard form, /stjudənt/).
- Because activation is graded rather than binary, this initially imperfect match doesn't derail recognition; the word can still climb toward recognition as further confirming input arrives.

Selection: Choosing the Winning Candidate

- If, as the Cohort Model claims, multiple word candidates get contacted and activated in parallel on the basis of a word's opening sounds, then some kind of selection process must ultimately choose between all of these competing candidates so that a single word can actually be recognized.
- **The Uniqueness Point**
- Definition: for any given word, we can identify its uniqueness **point- the specific point within the word at which it stops overlapping with all other words** that were originally part of its initial cohort. Past this point, no other cohort member remains a viable match.

- **Example:** Depending on exactly how the speaker pronounces it, for the word captain (/kæptən/), the uniqueness point could fall at the second vowel sound, /ə/, because this is the specific point at which captain finally becomes phonetically distinct from captive (/kæptɪv/). Up until that vowel, the two words are still indistinguishable from the input alone.

How uniqueness points are established/verified, methodologically:

- Theoretically, uniqueness points can be calculated in advance by searching through pronunciation-based dictionaries or lexical databases, to see exactly where a given word's sound sequence stops being shared with any other word.

- The unresolved question of mechanism- **HOW does the system "decide" a word has been recognized?** There has been genuine debate in the field about this:
  - One possibility: recognition occurs once a candidate word's activation level crosses some fixed threshold level.
  - Alternatively: recognition instead depends on there being a sufficient difference between the relative probabilities/activation levels of the various words remaining in the cohort- essentially, a competitive "gap" opening up between the front-runner and its closest rival (this can be thought of as the size of the difference between the activation curves shown in Figure 8.1).

Recognizing Nonwords: Does the Same Mechanism Apply?

- The puzzle: how do we manage to recognize, almost instantly, that a made-up string like splundle is not an actual word of our language?
- By definition, there is no stored representation in the mental lexicon corresponding to a word that doesn't exist- so what exactly is being "matched" or "rejected"?
- The claim made by the Cohort-style account: the recognition of nonwords is argued to follow exactly the same underlying process as the recognition of real words.
- This is described as an optimally efficient process, in the sense that decisions (whether "yes, this is a known word" or "no, it isn't") are made as rapidly as possible given the available information.

Why it makes theoretical sense that real words and nonwords would be processed identically:

- we, as listeners, have no way of knowing in advance, before we actually hear a given word, whether it is going to turn out to be a real word we know or a nonsense string, so we cannot possibly deploy some different, special-purpose strategy in advance of hearing it.
- The recognition system has to treat every incoming string the same way, right up until the evidence forces a particular outcome.

What actually happens, mechanistically, for a nonword:

- Exactly as with real words, a set of candidate words matching the initial portion of the input gets activated, and then, as more of the (nonsense) input is processed, the activation levels of these candidates change, shift, and decline, until eventually no remaining candidate word survives as a strong match.
- This point, where the very last viable real-world candidate finally drops out, occurs precisely at the point in the nonsense string where it diverges from every known word.
- This is called the **deviation point-** the equivalent, for nonwords, of the uniqueness point for real words. It marks where the input definitively stops matching anything stored in the lexicon.

Experimental Evidence for the Deviation Point

The founding experiment (Marslen-Wilson, 1980):

participants were required to press a response button as soon as they detected that they had just heard a nonsense (nonword) item within a list of stimulus words.

Result: response times, when measured starting from the deviation point (rather than from the start of the word), were found to be reasonably constant, at around 450 ms, across different nonword stimuli.

- Subsequent studies (Goodman & Huttenlocher, 1988; Taft & Hambly, 1986) found that response times measured from the deviation point are not perfectly constant, as all this variation depends on two factors:
  - The nature of the specific task the participants are performing.
  - The word-likeness of the material that comes after the deviation point in the nonword.
- It appears that listeners continue actively monitoring the incoming input even after the deviation point has technically already been passed.
- As a result, if a nonword happens to end in something that sounds more "word-like“, for example, a common, regular English affix/suffix such as -ise or -ic , this will noticeably slow down participants' "no, not a word" rejection responses, because that word-like ending creates some lingering uncertainty even after the core deviation point.